



OPO STARTUPS · JULY 2026

AI-Powered Hybrid Search & Agentic Assistants

on Cloudflare's Edge

Build production search and durable AI agents that scale to zero —
and back up.

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Senior DevOps Engineer · Applied Systems



Why I Built This — It Starts at the Workbench

Before the architecture, the problem. At **Juiced Wheels** a bench tech has a customer's e-bike on the stand and the clock running — and the answer is buried in a parts catalog and a stack of manuals no keyword box can navigate.

84K⁺

parts to search

The right innertube is one SKU among tens of thousands across QBP · JBI · OSCO.

2,845

PDF manuals

The fix is on page 47 of a Bafang PDF — not in a tidy, queryable database.

"30h?"

cryptic error codes

Bafang, Bosch & Shimano each speak a different dialect — the code is meaningless without the table.

≠

symptom vs. keyword

"Motor cuts out climbing" isn't a phrase in any manual — it's a trace through the system.

A bench tech doesn't want ten documents — they want **the answer, cited, in seconds**, and bench time is a shop's biggest cost. That's the bar generic search couldn't clear, so I built **Shop Shepherd** on Cloudflare for ~\$80/mo — and the stack underneath **isn't bike-specific**.

! The Problem with Traditional Search Infrastructure

⚠️ EVEN WHILE YOU READ THIS SLIDE...

\$0.0000

Burned by always-on infra
(*\$1,950/mo baseline, \$0.0007/sec*)

0

Queries actually served

100%

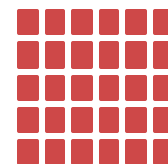
Of spend on idle capacity



Elasticsearch / OpenSearch

Dedicated clusters running 24/7 even with zero traffic.

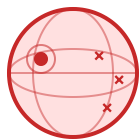
\$200–800+/mo minimum, no queries needed



Pinecone / Weaviate

Vector DB subscription bills for **all 30 days**, used or not.

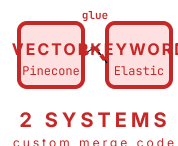
\$70–250+/mo serverless tier



Regional Lock-in

Pinecone, ES clusters pin to one region. EU/Asia queries pay **200ms+** tax.

Multi-region · 2–3× the cost or self-managed replication



Stitched Stack

Vector DB + keyword search are **separate systems**. Two bills, two query APIs, glue code to merge.

No native hybrid · sync drift · double the ops

CF Why Cloudflare? The Edge Advantage

✓ SAME SLIDE, ON CLOUDFLARE'S EDGE...

\$0.0000 —

Always-on cost

(flatline — pay only for what runs)

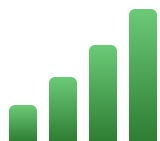
1,247,832

Queries served while you read

\$24.96

Total spend

(~\$0.00002 / query)



Usage-Based Pricing

Pay per request, per row, per query — not per hour.

Zero traffic = zero cost



Global Edge Network

Workers run within ms of the user, not in us-east-1.

330+ PoPs worldwide



Unified Platform

Workers + D1 + Vectorize + KV + R2 in one ecosystem.

One bill, no glue code



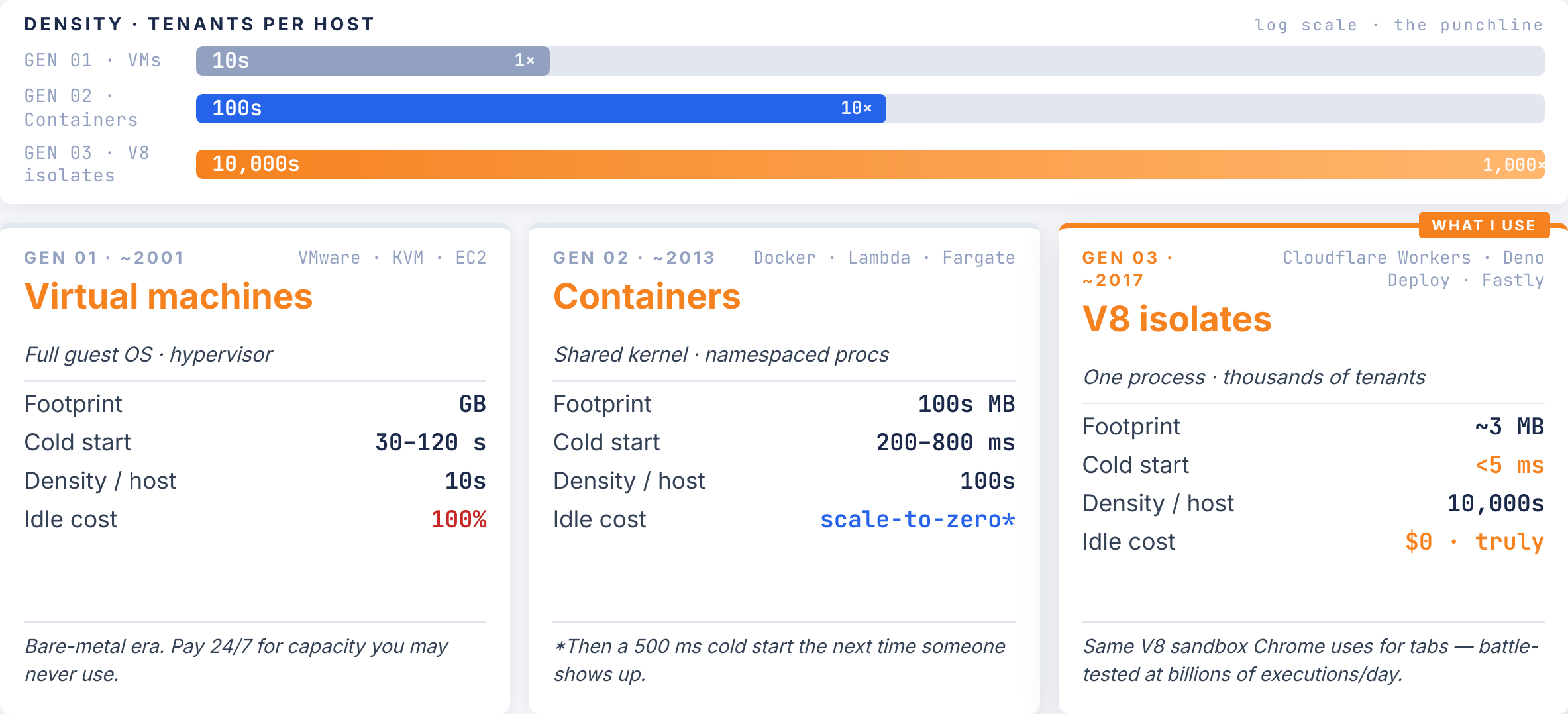
Zero Cold Starts

V8 isolates start in sub-ms — the shop lights flick on, no warming a cold engine.

200x faster than containerized FaaS

Three Generations of Compute

Each one cut cost and blast radius by an order of magnitude. V8 isolates are what makes scale-to-zero affordable at edge density.



PART ONE

Search Architecture Deep Dive

Three layers of search, one edge platform

\$ The Search Stack — Think of It Like Shopping

Three search engines, three shopping behaviors. The customer's voice picks the station: keyword for exact parts, semantic for symptoms, faceted for browsing with constraints.



Measured engine times, warm queries. [†]Semantic embed: ~3ms cached / ~200ms cold. [‡]Meili is remote-hosted (NYC); warm round-trip adds ~400-700ms — FTS5 + Vectorize are co-located in the worker.

|| Layer 1: Scan the Barcode — D1 + FTS5

LAYER 1 OF 3

How It Works

D1 is Cloudflare's serverless SQLite; FTS5 is its built-in full-text search — co-located with your Worker, no extra service, no extra bill.

FINDING · PROGRESSIVE RELAXATION

one round trip

EXACT	"BR-M6100" → 1 hit
PREFIX	"BR-M61..." → 6 hits
FUZZY	"BR~M6100" → 18 hits

RANKING · HOW BM25 SCORES A HIT

3 signals → one score

RARITY	"BR-M6100" outweighs a common "brake"
FREQUENCY	more hits help — but less each time
BREVITY	a title hit beats one deep in a manual

Exact-Match Lookup

CUSTOMER ASKS

"I need part BR-M6100"



BR-M6100

Shimano Deore Hydraulic Lever

BM25 score 8.4 · 7 ms (measured)

FTS5 STAGES

1 tokenize



2 match (virtual table)



3 rank (BM25)

~ Layer 2: Find the Right Aisle — Vectorize

LAYER 2 OF 3

Why Vector Search Matters

Keywords fail when users describe problems instead of naming parts. "My bike makes a clicking sound when I pedal" should match "bottom bracket adjustment" — vector search makes this connection.

Workers AI maps each sentence to **768 numbers** — coordinates on a "meaning map," where similar meanings land near each other even with zero shared words. **embeddinggemma-300m** generates them; Vectorize finds the nearest via cosine similarity.

Embedding Pipeline

- 1 Auto-chunk content > 7,500 chars
- 2 Workers AI → 768-dim vector
- 3 Vectorize index → stored at edge
- 4 KV cache → skip re-embedding

VECTORIZE STAGES

1 embed (Workers AI)



2 cosine search



3 topK=10

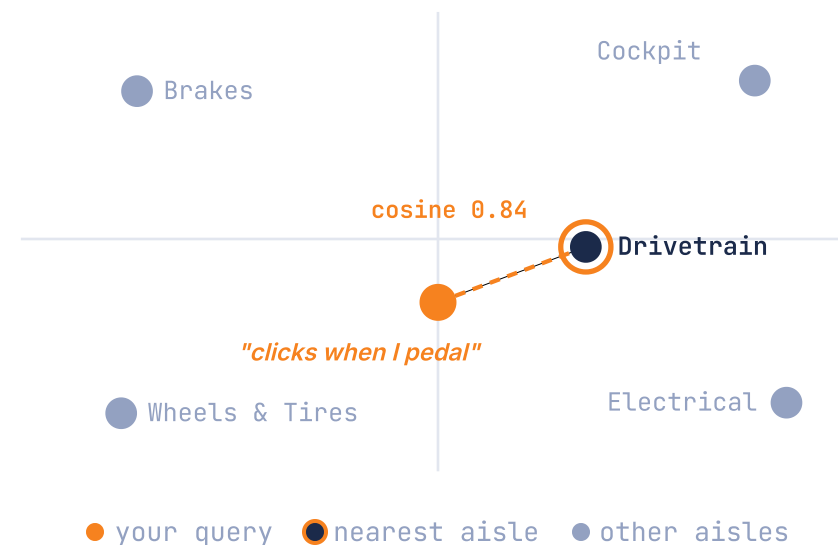
Semantic Routing

CUSTOMER ASKS

"My bike clicks when I pedal"



MEANING MAP • NEAREST AISLE WINS





Layer 3: Pick Off the Shelf — D1 + Meilisearch

What Faceting Is

Filtering a big catalog by structured attributes — brand, wheel size, valve type, price, in-stock — with a **live count on every option**. It's how a tech narrows 84,000 parts down to the 8 that fit, without typing a perfect query.

Why D1 *and* Meilisearch?

D1 system of record

Normalizes 84K parts from 3 vendor feeds, scores e-bike relevance, and serves keyword search via FTS5 — the fallback if Meili is ever down.

Meilisearch the speed layer

A sub-30ms count for every facet checkbox across all 84K rows — which a D1 `GROUP BY` can't keep up with — plus typo tolerance: "shrader" still finds "Schrader". Synced from D1.

Faceted Shelf in Action

CUSTOMER ASKS

"Innertube — 26" × 4", Schrader valve"

FILTERS APPLIED

☒ Tire width: 4.0"	84
☒ Wheel diameter: 26"	46
☒ Valve type: Schrader	24
☒ In stock	8



26" × 4.0–4.8" Fat Innertube

Kenda · 35mm Schrader valve · \$13.99

typo-tolerant · ~28ms · fallback to D1 FTS5

THE LONE THIRD-PARTY

Meilisearch (\$30/mo) is the only non-Cloudflare piece of this whole stack — today's one platform gap for fast faceting + typo tolerance. If Cloudflare ships a native equivalent, this line drops to **\$0** and the architecture is 100% first-party.

MEILISEARCH STAGES

1 parse facets



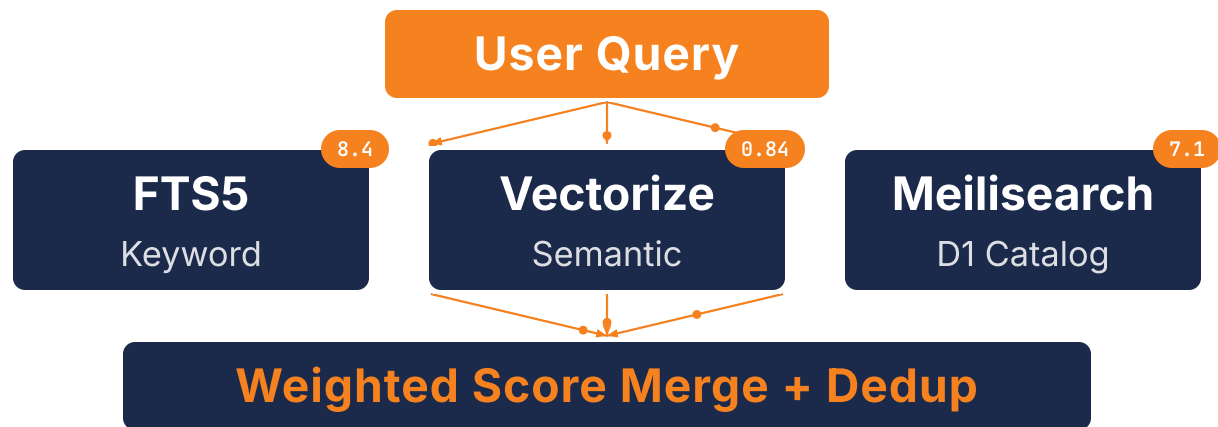
2 filter (typo-tolerant)



3 rank by relevance

⊕ Hybrid Search: Merging Keyword + Vector

When one engine isn't enough, fire all three in parallel and merge by weighted relevance score. The agent never sees three result lists — just one ranked output.



TOP RESULT | **BR-M6100** Shimano Deore Hydraulic Lever — matched by all three engines

final 18.2

Source Weights

Manuals 2.0× · Docs 1.5× · Parts 0.2×

- OEM manuals win every tie — they're the authoritative source for technicians
- Parts get demoted unless the query is clearly a SKU lookup
- A result returned by multiple engines stacks the boosts

Short Query Optimization

1-2
words

70% keyword

30%

3+
words

50% keyword

50% vector

- Short queries lean keyword — prevents false matches like *"venue"* ≠ *"venus"*
- Below-floor results get cut, with a top-3 fallback so the agent never starves

PDF Pipeline: OCR, Extract, Optimize

Why OCR Every PDF?

Most OEM service manuals are scanned images or flattened exports — there's no selectable text layer. **OCR (Optical Character Recognition)** turns those page images into searchable text. I then inject that text **back into the PDF as a hidden layer**, so the manual is fully searchable in any viewer while looking identical to the original. **You can't chunk what you can't read.**

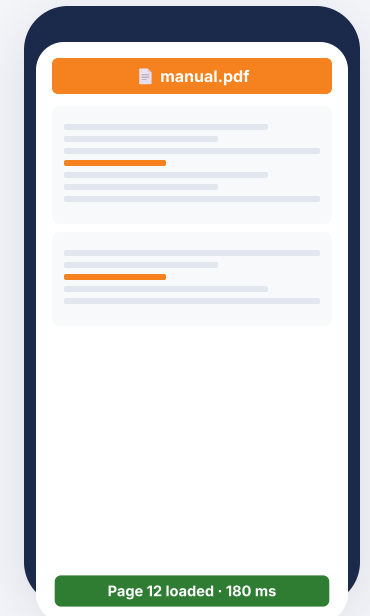
RAW OEM PDF → MOBILE-READY



40× smaller · same content · ready for a 4G phone at the bench

HOW I GET THERE

- ✓ Subset embedded fonts (*strip unused glyphs*)
- ✓ Recompress images (*JPEG2000 / lossy DCT*)
- ✓ Inject OCR text layer (*hidden, selectable, indexable*)
- ✓ Optimize via Adobe to PDF/A-2u (*latest archival standard*)
- ✓ Web-linearize (*first page renders before full download*)



Tech reads at the bench

PDF STAGES

1 OCR



2 chunk + embed



3 D1 + Vectorize



4 R2 (mobile + thumbs)



Why Chunking Context Matters

The Naive Chunking Problem

Most tutorials split documents into fixed-size blocks and embed them as-is — like cutting a manual into unlabeled index cards. You get a match back, but no idea which card it's from.

An orphaned chunk like *"torque the bolts to 40 Nm"* is useless without knowing it's a Bosch CX mid-drive install vs. a Shimano hub motor teardown.



Long manual PDF · 142 pages

Bosch CX Mid-Drive Install Guide

↓ chunked + tagged

chunk_0042

"Torque bolts to 40 Nm. Check alignment..."

bosch-cx install

chunk_0043

"Speed sensor mounting: 5 Nm. Use anti-seize..."

bosch-cx sensor

chunk_0044

"Battery seat alignment: shim to 0.2 mm..."

bosch-cx battery

Naive Chunk (no context)

```
{
  text: "Torque bolts to 40 Nm.
        Check alignment before
        tightening...",
  vector: [0.23, -0.41, ...]
}
```

Metadata-Tagged Chunk (structural context)

```
{
  text: "Torque bolts to 40 Nm...",
  vector: [0.23, -0.41, ...],
  metadata: {
    title: "Bosch CX Installation",
    path: "/guides/bosch-cx-install",
    tags: ["mid-drive", "torque"],
    category: "installation",
    source: "manual"
  }
}
```

NEXT

Metadata fixes naming. But what about OCR garbage where even the chunk text is scrambled? →

Contextual Retrieval: Quality Up, Cost Down



Metadata gives chunks a structural home. A 1–2 sentence **LLM summary** — Anthropic's Contextual Retrieval pattern — adds a sticky note that stays findable even when the page is OCR garbage.

Labor division: *LLMs author chunk context; humans author the graph (slides 19–21) — enrich text, don't invent ontology.*

BEFORE — RAW OCR CHUNK

Indexes can't find it

```
"340 % 8 1 240% TURBO 340%
3 4 0 SPEED SENSOR BDU
4xx INSTALL torque 5Nm..."
```

- ✖ **FTS5 BM25:** 0 useful matches — query terms don't appear in garbled OCR
- ✖ **Vectorize:** embedding of garbage = garbage similarity

Query: "Bosch Drive Unit speed sensor installation"

AFTER — CHUNK + context_llm

Both indexes hit at rank 1

context_llm
Bosch DriveUnit BDU4xx (MY24) installation guide — speed sensor mounting and torque spec.

```
"340 % 8 1 240% TURBO 340%..."
```

- ✔ **FTS5 BM25:** "Bosch", "DriveUnit", "speed sensor" all match the summary
- ✔ **Vectorize:** coherent summary embeds cleanly → high cosine sim

RETRIEVAL QUALITY · n=30 cases

Top-1 accuracy

66.7% → **93.3%** +26.6 pp

Top-3 accuracy

83.3% → **100%** +16.7 pp

COST PER TURN

Dollars per turn

\$0.201 → **\$0.075** -63%

Step-2 input tokens

49.6K → **13K** -73%

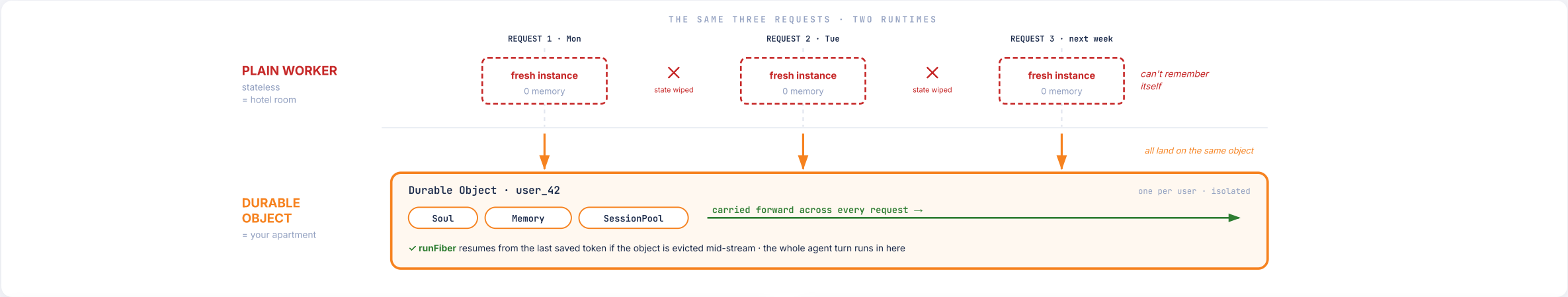
PART TWO

From RAG to Agentic AI

Everything so far was stateless — every request started cold. Now I add what search can't have: durable per-user memory, crash recovery, and query-aware context compression. Watch one number the whole way down: a tool turn drops from ~36¢ to ~3¢.

A Durable Object Per User (Project Think)

A Worker is a **hotel room** — fresh every request, your stuff is gone. A Durable Object is your **apartment** — same address, same key, your memory is still on the counter.



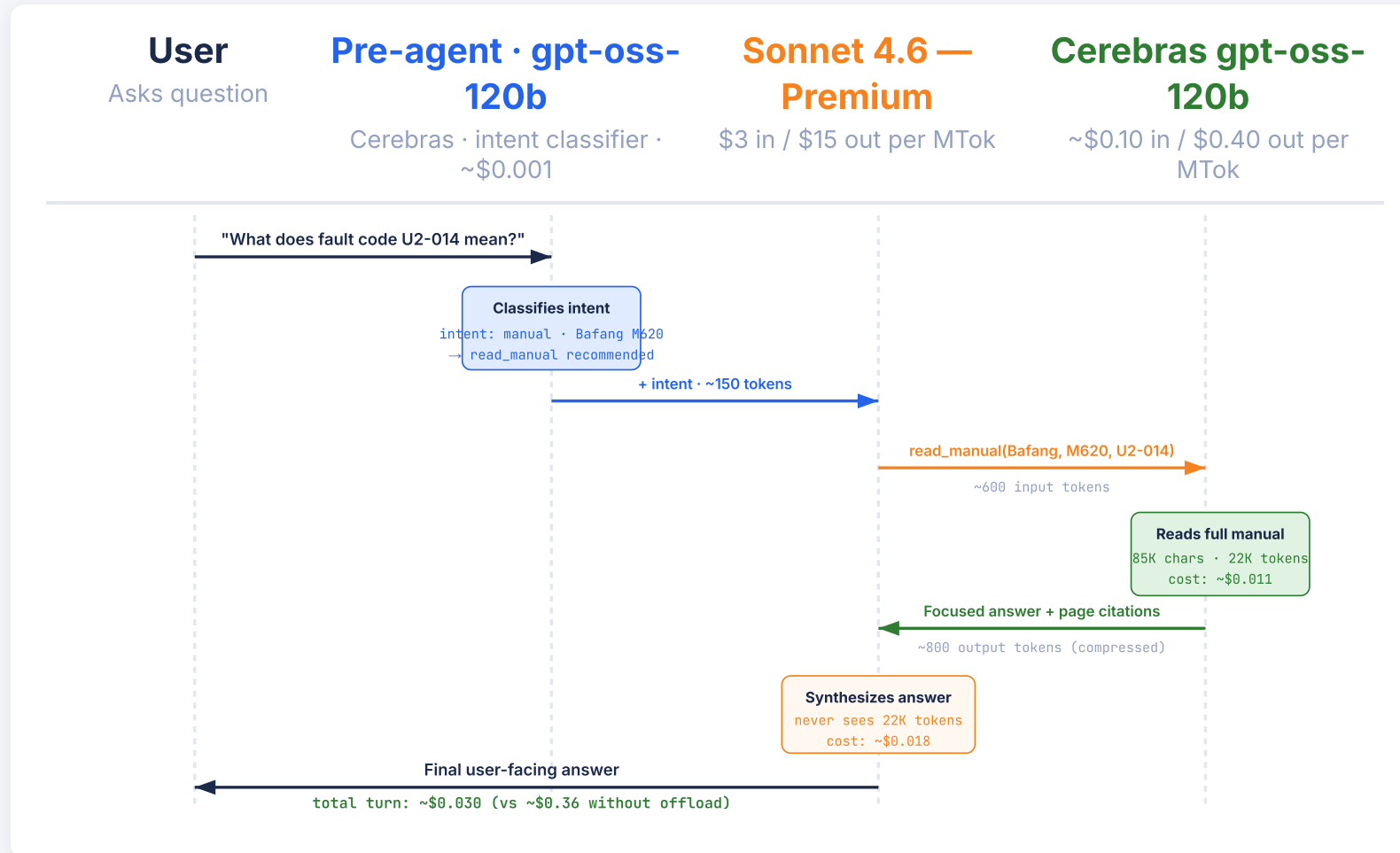
TOOL	PURPOSE
search_documentation	unified hybrid search
search_manuals	PDF manual sections
search_parts	faceted parts catalog
query_ebike_graph	upstream / downstream / interrupts
set_context · find_symptom	user memory + symptom search

PER-USER DURABLE OBJECT STATE
Soul block System prompt from KV — live-refresh per turn
Memory block Per-user facts via set_context — durable identity
SessionPool Per-thread history, FTS5-searchable
chatRecovery runFiber — mid-crash saves partial text

↔ Sub-Agent Offloading: Cheap Bulk Work, Premium Synthesis

High-token tool work goes to a near-free model; the premium model only sees the compressed result and writes the answer. Like a shop, the apprentice does the bulk grunt work and the master tech makes the call.

~\$0.36 → ~\$0.030 per turn — a 90% cut, same answer quality.



THROUGHPUT TOKENS / SECOND

Cerebras gpt-oss-120b (used here)

~3,000 t/s

20–60× faster

Groq · Together (specialized inference)

~500–1,000 t/s

~7×

Sonnet 4.6 · GPT-5 · Gemini 2.5 (frontier APIs)

~50–150 t/s

1× baseline

20–50× faster than frontier APIs.

The pre-agent adds ~250ms, not ~3 seconds — that's what makes the offload pattern viable.

Why Cerebras for the mini-agent: I trialed Claude Haiku and Cloudflare Workers AI models too — both viable, but Cerebras's speed at low cost was the best fit for work that runs on every turn.



Context Engineering: The Missing Layer

UNIT ECONOMICS

× BEFORE · DUMB TRUNCATION

The model gets noise.

Like clipping the manual at page 16.

- Tool returns **15 KB** raw PDF text
- Char-count clip at 16K characters
- Sonnet sees 8 irrelevant pages of OEM filler
- Burns **8 follow-up tool calls** narrowing down
- No page citations, no section anchors

Response: 230 chars of vague advice

✓ AFTER · QUERY-AWARE COMPRESSION

The model gets answers.

Like a service writer marks the right page.

- Same 15 KB result — pre-Sonnet hop adds `gpt-oss-120b`
- Cerebras sub-agent extracts query-relevant sections
- **Page citations + chunk IDs preserved**
- Sonnet receives **3 KB** of focused content
- Answers in **2 calls** with diagnostic depth

Response: 2,239 chars of specific guidance

INPUT TOKENS

~~15K~~ → **3K**

per tool result sent to Sonnet

-80%

TOOL CALLS / ANSWER

~~8~~ → **2**

no follow-up narrowing

-75%

COST / TURN

~~\$0.36~~ → **\$0.03**

≈ \$330 saved / 1K turns

-90%

USEFUL RESPONSE

~~230~~ → **2,239**

chars of specific guidance

+10×

RAG retrieves. Prompting reasons. Compression curates.

Always-compress tools: `read_manual` · `get_manual_content` · `get_content_section` · others compress only above the 4K-token budget

When retrieval *isn't enough*.

Search returns **text**. Diagnosis needs **causality** — what supplies what, what cuts what, what symptom traces to which subsystem.

01 WHAT I HAD

Hybrid search over 2,845 manuals, 2,334 articles, 84K parts.

- FTS5 + Vectorize + Meilisearch fused via `/api/search/unified`
- Weighted merge + **Gemini structural rerank** — ranked passages with page citations

Great at finding the **page**. Useless at finding the **cause**.

02 WHAT BROKE

"Motor cuts out climbing" returned 12 pages.

- Pages described the **symptom** (motor cutting out under load)
- None named the **cause** (controller overheating) or the **subsystem** (thermal cutoff disabling motor power)

A symptom isn't a keyword. It's a **trace** through the system.

03 WHAT I BUILT

A diagnostic graph built on physics, real-world conditions & rider behavior.

- 825 profiled nodes — *each carries tools · symptoms · tiered \$0-first steps*
- 2,523 typed edges — causal verbs the agent calls per turn via `query_ebike_graph`

SUPPLIES_POWER_TO · CUTS_POWER_TO ·
SYMPTOM_OF · SENDS_SIGNAL_TO · TESTED_BY

NEXT THREE SLIDES

I tried LLM extraction. It hallucinated. I shipped hand-crafted instead — and the agent still queries the graph every diagnostic turn. Here's the trade-off, the anatomy, and the live trace.

Why *hand-crafted* beat LLM extraction.

LLM EXTRACTION · TRIED IT

High cost. High risk.

DEAD-END

- 963 docs → Anthropic Batch API → entities + relationships
- 31 duplicate "Battery" variants → canonicalization nightmare
- No auto-sync — graph drifts every time a doc changes
- Hallucinated relationships (`motor PART_OF tire?`)
- ~\$40 per full re-extraction × every doc update
- No audit trail — can't diff a node's history

Kept it OFF in production



HAND-CRAFTED · SHIPPED IT

\$0 cost. 100% deterministic.

SHIPPED

- 825 nodes in TypeScript — one file, one source of truth
- Expert-authored with e-bike diagnostic ordering
- Regenerate SQL + seed in seconds — zero LLM cost
- Zero hallucinated edges → no fabricated repair advice
- Every change is a PR with a reviewable diff
- Agent still uses GraphRAG to `traverse` — humans author, LLMs query

LIVE in production, every diagnostic query



” Truth is human work. Reasoning over it is LLM work. ”

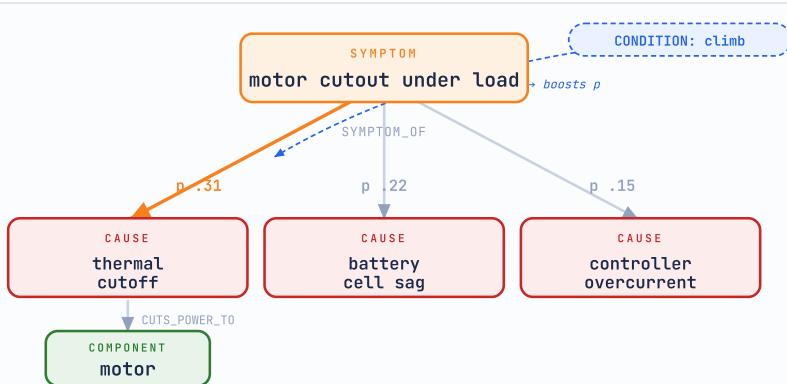
INTEGRATED WITH EVERY DIAGNOSTIC QUERY

```
find_symptom('motor cutout climbing') → sym_motor_cutout_under_load (8 ms · D1)
query_ebike_graph('controller', 'interrupts') → brake_cutoff_sensor, shift_sensor (safety overrides)
lookup_error_code('E021') → controller_thermal_cutoff (69 codes indexed)
```

Not a search index — a **diagnostic brain**.

825 expert-authored diagnostic profiles · 2,523 weighted causal edges — *error codes are just one way in.*

ONE DIAGNOSTIC SLICE · D1 + KV



A real slice of the 825-node graph. **Typed nodes** (symptom · cause · component) linked by **typed, weighted edges** — and live **conditions** (a sustained climb) re-weight the probabilities before ranking. **Expert-authored, not LLM-scraped.**

symptom cause / failure component edge = relation · **p** = probability

ALSO IN THE GRAPH

18 node types · 85 typed relations · environmental multipliers (rain · cold · cargo)

SYMPTOM_OF · CAUSES_FAULT_IN · FAILS_AS · TESTED_BY · INTERRUPTS · CUTS_POWER_TO

A PLAYBOOK PER NODE — thermal_cutoff

TOOLS	multimeter · IR thermometer · torque wrench
CONSUMABLES	dielectric grease · thermal paste
SYMPTOMS	cutout under load · power returns after cooldown

TIER 0 · \$0	temp check after a climb · airflow + brake drag
TIER 1 · meter	temp-sensor resistance · connector continuity
TIER 2 · swap	replace thermal sensor / controller

Every node carries this — cost-tiered, e-bike-specific, cheapest-first.

HOW GRAPHRAG ANSWERS — "motor cuts out climbing"

- 1 MATCH** `find_symptom` maps the phrase to a **node** — codes & the 768D vector are other ways in → `sym_motor_cutout_under_load`
- 2 TRAVERSE** `query_ebike_graph` walks causal edges to pull the connected subsystem
- 3 RANK** scored by base $p \times$ live conditions (climb = load + heat) → `thermal_cutoff` · `battery_cell_sag` · `controller_overcurrent`
- 4 ANSWER** Sonnet reasons over the **causal subgraph** and names the cause + the fix

Plain RAG retrieves *passages*.

GraphRAG retrieves the **causal chain**.

From symptom to *diagram* — live.

Shop Shepherd · shepherd.shopdawg.app

claude-sonnet-4-6

Customer says the motor cuts out when climbing hills. Bafang M620.

→ tool: find_symptom

```
{ "query": "motor cutout climbing", "limit": 3 }
```

← sym_motor_cutout_under_load 12 ms · D1

→ tool: query_ebike_graph

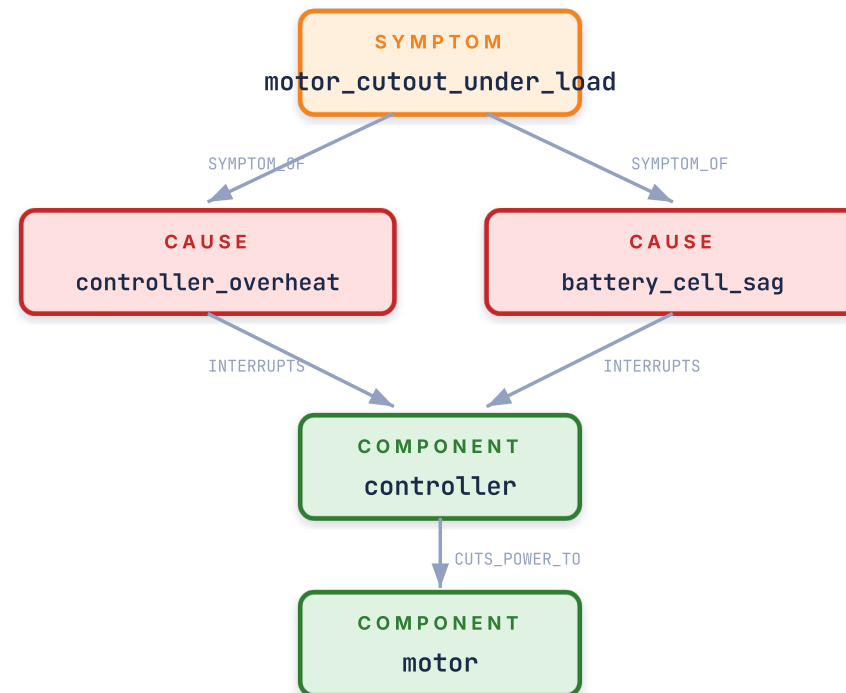
```
{ "node": "controller", "verb": "interrupts" }
```

← 4 nodes, 5 edges · rendering trace... 8 ms · D1

Likely **controller overheating** or **battery cell sag** — under sustained climbing load, pack voltage drops or the controller overheats, and its cutoff trips. Diagram on the right.

GENERATED MERMAID · RENDERED INLINE

graph TD



■ symptom

■ cause

■ component

→ LIVE DEMO

IF YOU DON'T WORK ON BIKES **motor** = the web server · **controller overheating** = CPU throttling under load ·

battery sag = DB / resource latency under load



Real-World Scale: What This Handles

2,845



PDF Manuals Indexed

Gathered with AI search — Brave · Tavily · Firecrawl
OCR'd · ~593K chunks · full-text + vector indexed

2,334



Repair Articles

Built from Google Deep Research, human-curated
full-text + vector indexed

84K+



Parts in Catalog

Nightly refresh from 3 vendor feeds
D1 faceted, Meili-accelerated

\$80/mo



Production Stack Cost

Workers + D1 + Vectorize + Meili included

\$

Cost Comparison: Traditional vs. Cloudflare

UNIT ECONOMICS

WHERE THE MONTHLY SPEND GOES · REPRESENTATIVE PRODUCTION SCALE · JUN 2026 RATES



CLOUDFLARE'S \$80, ZOOMED IN 25× — WHERE IT ACTUALLY GOES



Orange = Meilisearch, the only paid third-party — skip it and D1 FTS5 covers parts (Workers Paid \$5 · D1 free tier).

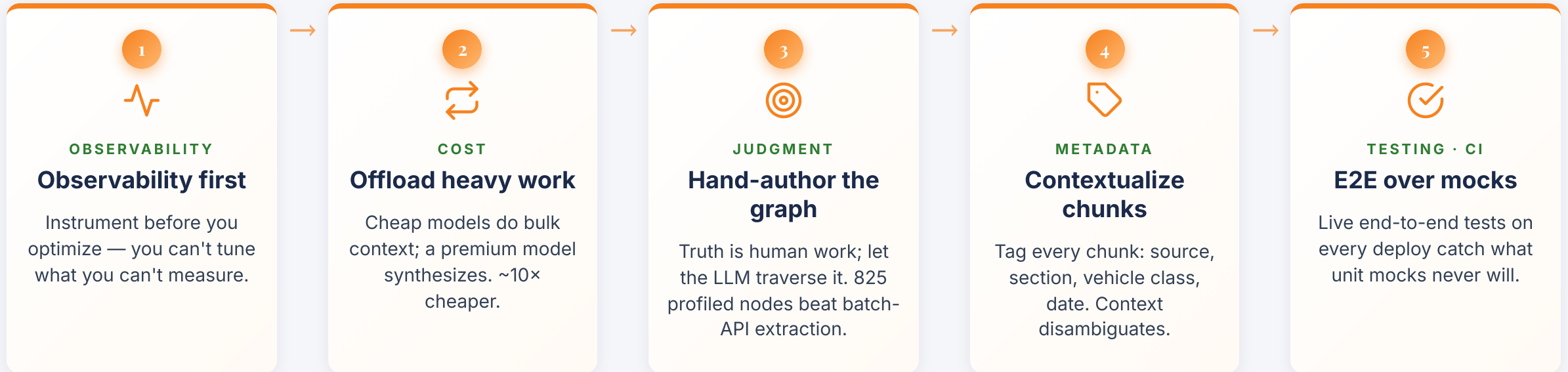
25× cheaper · same throughput · scale-to-zero baseline





Lessons from Production

Five hard-won lessons from running this in production — **not the order I learned them, but what I'd hand my past self on day one.**



THE ONE-LINE VERSION

Let **cheap models do the volume**, keep **humans on the judgment calls**, and **measure everything** in between.

PART THREE

Build It Yourself

Two steps, not two choices — ship on AI Search now, graduate to the primitives when you scale.

None of it is bike-specific — swap in your industry's manuals, parts & diagnostics and the architecture holds.

Cloudflare *AI Search*.

The same architecture you just saw — provisioned for you in a few clicks.

WHAT YOU GET OUT OF THE BOX

R2	Drop PDFs, HTML, Markdown into a bucket. That's the ingest.
Workers AI	Embeddings generated automatically — same models you'd wire up by hand.
Vectorize	Index provisioned and kept in sync with R2 changes.
AI Gateway	Logging, caching, retries on every model call — included.
Search API	One endpoint — ranked results or a streamed answer with citations.

IF YOU REMEMBER ONE THING

Same R2 + Workers AI + Vectorize + AI Gateway pipeline.

Click to provision. No wrangler glue.

Drop docs into a bucket. Auto-chunked, auto-reindexed.

Hit one endpoint. Ranked results or full RAG answer.

Same scale-to-zero economics. Same edge latency.

developers.cloudflare.com/ai-search

START HERE

If you don't need per-tenant Durable Object state, hand-tuned hybrid scoring, or a long tail of agent tools — **AI Search ships you 80% of this deck without writing a Worker** — and every primitive is right here in Step 2 when you outgrow it.

Same primitives, *full control*.

1

DATABASE · SQL

D1 + FTS5 for keyword

```
wrangler d1 create my-search-db
```

```
CREATE VIRTUAL TABLE ... USING fts5()
```

2

VECTORS

Vectorize for semantic

```
wrangler vectorize create my-index \
```

```
--dimensions=768 --metric=cosine
```

3

INGEST · COMPUTE

Worker to ingest & embed

Chunk with metadata, embed via Workers AI, write to D1 + Vectorize in one transaction.

4

SEARCH · API

Hybrid endpoint — merge & score

Parallel FTS5 + Vectorize, weighted merge by source, threshold + dedup.

5

AGENT · MEMORY

Agent layer for tools & memory

Install [@cloudflare/think](#) — streaming, tools, per-user Durable Object memory, crash recovery.

**WHEN YOU'RE
DONE**

You ship **hybrid search**, a **durable agent**, **per-user memory**, and **crash recovery** — every primitive owned by you, none in a vendor's stack.



Start Building Today

Everything shown here runs on Cloudflare's free or pay-as-you-go tiers

Start on AI Search → graduate to the primitives as you grow. Same edge, same economics, no rip-and-replace.

developers.cloudflare.com/d1

developers.cloudflare.com/vectorize

developers.cloudflare.com/workers-ai

cloudflare.com/startups

[blog: AI Search primitive](#)

[blog: Project Think](#)



SCAN FOR
THESE SLIDES

See it live → shepherd.shopdawg.app

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